

CNN

December 12, 2025

```
[33]: import numpy as np  
import matplotlib.pyplot as plt  
import pandas as pd
```

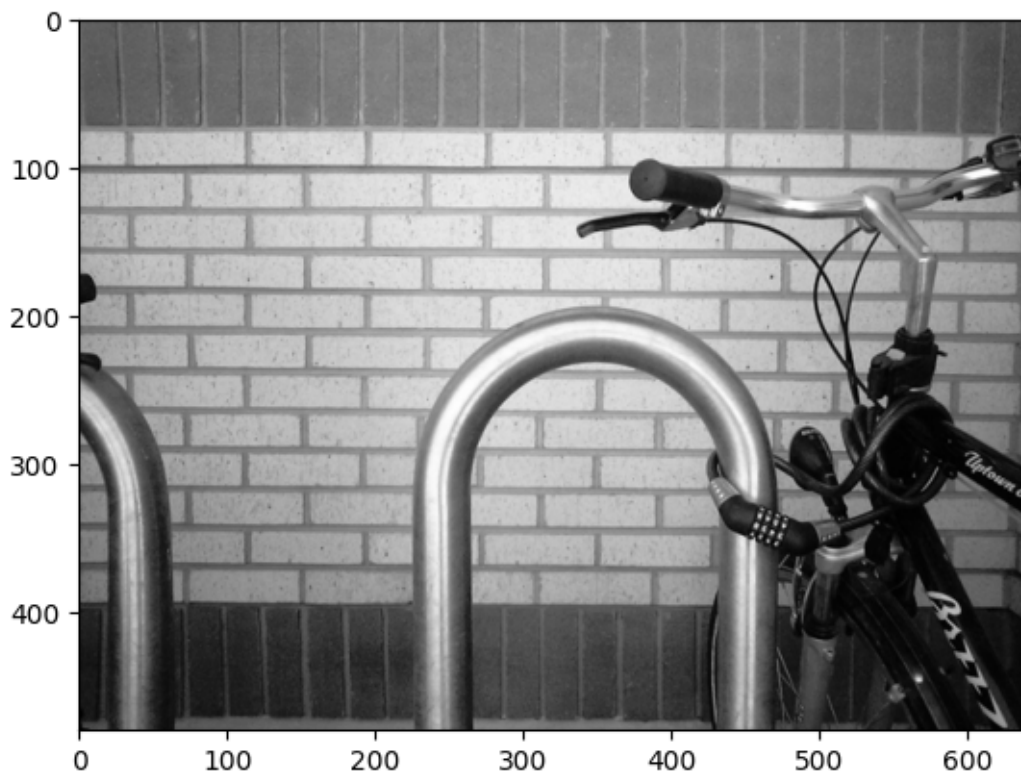
```
[34]: img = plt.imread("Bikesgray.png")
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[35]: img.shape
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[35]: (480, 640)
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[36]: plt.imshow(img, cmap = "Greys_r")
```

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[36]: <matplotlib.image.AxesImage at 0x299bdd6d9d0>
```



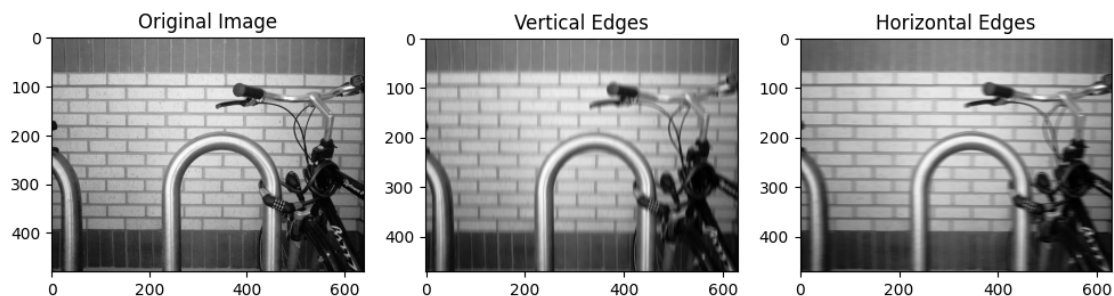
```
[37]: def apply_filter(image,filter):
    h,w = image.shape
    fh,fv = filter.shape
    newimage = np.zeros((h - fh + 1, w - fv + 1))
    for i in range(h - fh + 1):
        for j in range(w - fv + 1):
            newimage[i,j] = np.sum(image[i:(i + fh),j:(j + fv)] * filter , axis_
↵= (0,1))
    return newimage
```

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[38]: filter_size = 11
filterv = np.zeros((filter_size,filter_size))
filterh = np.zeros((filter_size,filter_size))
filterv[:,filter_size//2 + 1] = 1
filterh[filter_size//2 + 1,:] = 1
```

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[39]: imgv = apply_filter(img,filterv)
imggh = apply_filter(img,filterh)
```

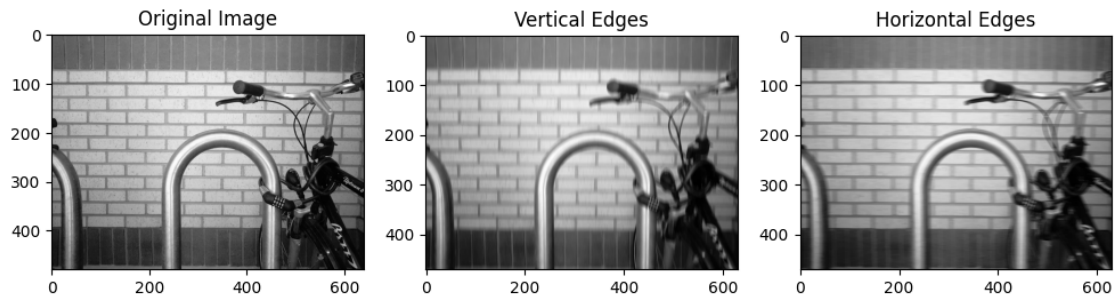
```
[40]: def plot(img,imgv,imggh):
    plt.figure(figsize = (12,6))
    plt.subplot(1,3,1)
    plt.imshow(img,cmap = "Greys_r")
    plt.title("Original Image")
    plt.subplot(1,3,2)
    plt.imshow(imgv,cmap = "Greys_r")
    plt.title("Vertical Edges")
    plt.subplot(1,3,3)
    plt.imshow(imggh,cmap = "Greys_r")
    plt.title("Horizontal Edges")
    plt.show()
```

```
[41]: plot(img,imgv,imggh)
```



```
[42]: imgv = apply_filter(img,filterv)
      imgh = apply_filter(img,filterh)
```

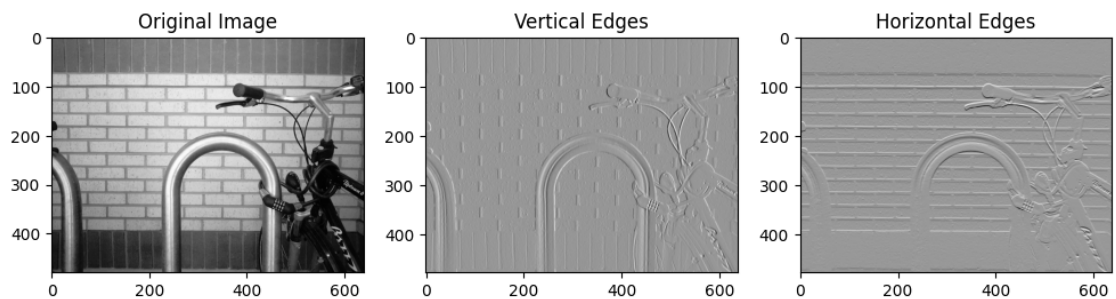
```
[43]: plot(img,imgv,imgh)
```



```
[44]: sobelv = np.array([[-1,0,1],
                        [-2,0,2],
                        [-1,0,1]])
      sobelh = np.array([[1,2,1],
                        [0,0,0],
                        [-1,-2,-1]])
```

```
[45]: imgv = apply_filter(img,sobelv)
      imgh = apply_filter(img,sobelh)
```

```
[46]: plot(img,imgv,imgh)
```



```
[51]: import os
      os.environ["KERAS_BACKEND"] = "torch"
      import keras
      from keras import ops
```

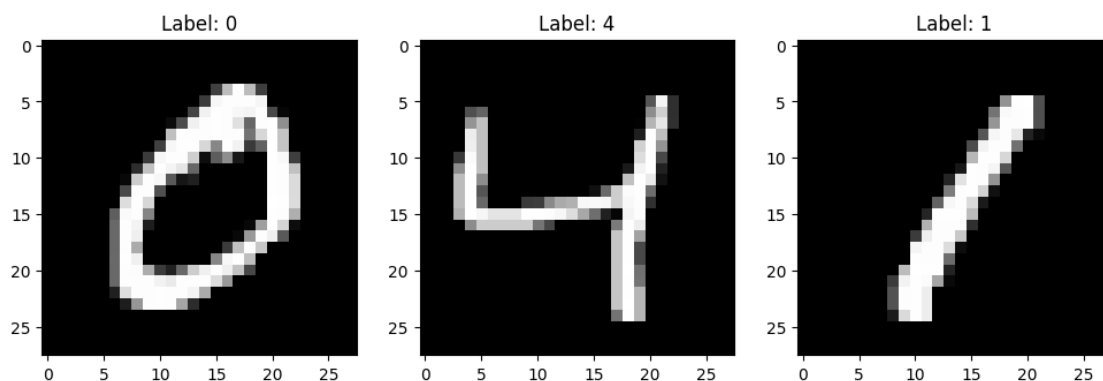
```
[ ]: trainData = pd.read_csv("mnist_train.zip")
testData = pd.read_csv("mnist_test.zip")
Xtrain , Ytrain = trainData.iloc[:,1:].to_numpy(),trainData.iloc[:,0].to_numpy()
Xtest , Ytest = testData.iloc[:,1:].to_numpy(),testData.iloc[:,0].to_numpy()
Xtrain,Xtest = Xtrain.reshape(-1,28,28),Xtest.reshape(-1,28,28)
Xtrain = Xtrain.astype("float32")/255
Xtest = Xtest.astype("float32")/255
```

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[58]: Xtrain.shape
```

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[58]: (59999, 28, 28)
```

```
[ ]: from keras.utils import to_categorical
Ytrain = to_categorical(Ytrain, 10)
Ytest = to_categorical(Ytest, 10)
```

```
[ ]: plt.figure(figsize = (12,4))
for i in range(3):
    plt.subplot(1,3,i+1)
    plt.imshow(Xtrain[i],cmap = "Greys_r")
    plt.title(f"Label: {np.argmax(Ytrain[i])}")
plt.show()
```



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[ ]:
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